CS1005 Discrete Structuress

BS CS Spring 2026

Final Project

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Discrete Structures Project:

Item-Level Price Co-Movement Network using CPI Data

Technical Report

Course: Discrete Structures

Data Source: Pakistan Bureau of Statistics (pbs.gov.pk)

YouTube Demo: [Insert YouTube Link Here]

1. Introduction

Consumer prices do not move on their own. When energy costs go up, food processing gets more expensive too. When seasonal produce fills the market, vegetable prices fall in ways that often copy each other across cities. The question this project tries to answer is: which consumer items tend to move together in price, and does that pattern stay the same from year to year?

This project builds a network where every node is a consumer item and every edge means that two items had similar price movement patterns across multiple cities. We use three years of monthly Consumer Price Index data from five major Pakistani cities: Karachi, Lahore, Islamabad, Peshawar, and Quetta. The data covers 51 items from seven product categories: Food, Energy, Clothing, Housing, Education, Health, and Transport.

By using cosine similarity on price change vectors and then applying graph theory tools such as degree centrality, closeness centrality, and betweenness centrality, we can find out which items are most central to the price co-movement network, which categories tend to cluster together, and how these relationships change from 2022 to 2023 to 2024.

2. Data Description

The dataset was downloaded from the Pakistan Bureau of Statistics website (pbs.gov.pk) and covers monthly retail prices for 51 consumer items across five cities over three years: 2022, 2023, and 2024. After loading and cleaning, the dataset has 9,180 records in long format.

Attribute	Details
Cities	Karachi, Lahore, Islamabad, Peshawar, Quetta
Items	51 consumer items
Categories	Food, Energy, Clothing, Housing, Education, Health, Transport
Years	2022, 2023, 2024
Frequency	Monthly (12 months per year)
Total records	9,180 after reshaping to long format

Each original row in the CSV had prices for all 12 months in a single row. We changed this into a long format so that each record has: City, Item, Category, Year, Month, and Price. Missing values were filled using forward-fill then backward-fill within each City-Item-Year group.

3. Methodology

3.1 Price Change Vectors

Raw prices have very different price levels depending on the item. A loaf of bread costs around 80 rupees while petrol costs over 200 rupees. Comparing raw prices directly would be misleading. Instead, we work with the monthly price changes.

For each item i, city c, and year y, we calculate an 11-element price change vector by taking the difference between each month and the previous month:

v(y)_i,c = (delta_p_2 , delta_p_3 , ... , delta_p_12)

where delta_p_m = price in month m minus price in month m-1. This gives us one vector per item per city per year, each of length 11.

3.2 Cosine Similarity

We measure how similar two items are within a given city using cosine similarity. If item i has change vector v_i and item j has change vector v_j, their similarity is:

sim(i, j) = (v_i dot v_j) / (||v_i|| * ||v_j||)

A cosine similarity close to 1 means both items tend to rise and fall by similar amounts in the same months. A value close to 0 means their price movements are unrelated. A negative value means they tend to move in opposite directions.

3.3 City Aggregation

Because we have five cities, we can check whether the similarity between two items is the same across Pakistan or only in one place. For a given year y and similarity threshold tau, we count how many cities show similarity at or above tau for the pair (i, j):

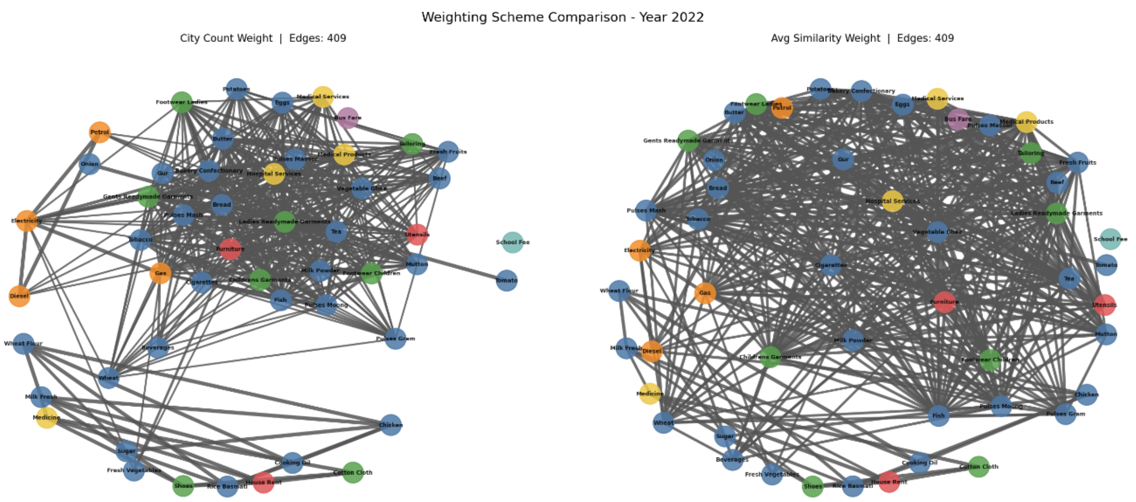
N_y(i, j) = | { c in C : sim(i, j) in city c >= tau } |

3.4 Graph Construction

For each year y, we build an undirected graph G_y where nodes are items and an edge exists between items i and j if N_y(i, j) is greater than or equal to K, a city-count limit set by the user. In this report we use tau = 0.7 and K = 2 as the main parameter setting.

We also build a weighted version of the graph using two weighting options:

- Option 1 - City Count Weight: the edge weight equals the number of cities that have similarity above tau. Higher weight means more cities agree on the relationship.
- Option 2 - Average Similarity Weight: the edge weight is the average cosine similarity across all five cities. This captures strength of correlation, not just count.



3.5 Centrality Measures

For each yearly graph we calculate three common centrality measures using the NetworkX library:

Measure	Formula Basis	Interpretation
Degree Centrality	Fraction of nodes connected	How many other items share a co-movement relationship with this item
Closeness Centrality	Inverse of average shortest path	How quickly a price shock spreads from this item to all others in the network
Betweenness Centrality	Fraction of shortest paths passing through node	How much a single item acts as a bridge connecting otherwise disconnected groups

4. Implementation

The entire pipeline is written in Python. The main libraries used are Pandas for data handling, NumPy for number operations, scikit-learn for cosine similarity calculation, NetworkX for graph building and centrality analysis, and Matplotlib for visualization.

The code is organized as a single file called main.py and is divided into clearly labeled functions for each step. The pipeline runs from start to finish, from loading the CSV to saving all output graphs as PNG files. Below is an overview of the main functions:

Function	Purpose
load_data()	Reads CSV, reshapes to long format, fills missing values
compute_change_vectors()	Calculates 11-element monthly price change vector for each city-item-year combination
compute_similarity_per_city()	Calculates pair-by-pair cosine similarity for all item pairs within each city for a given year

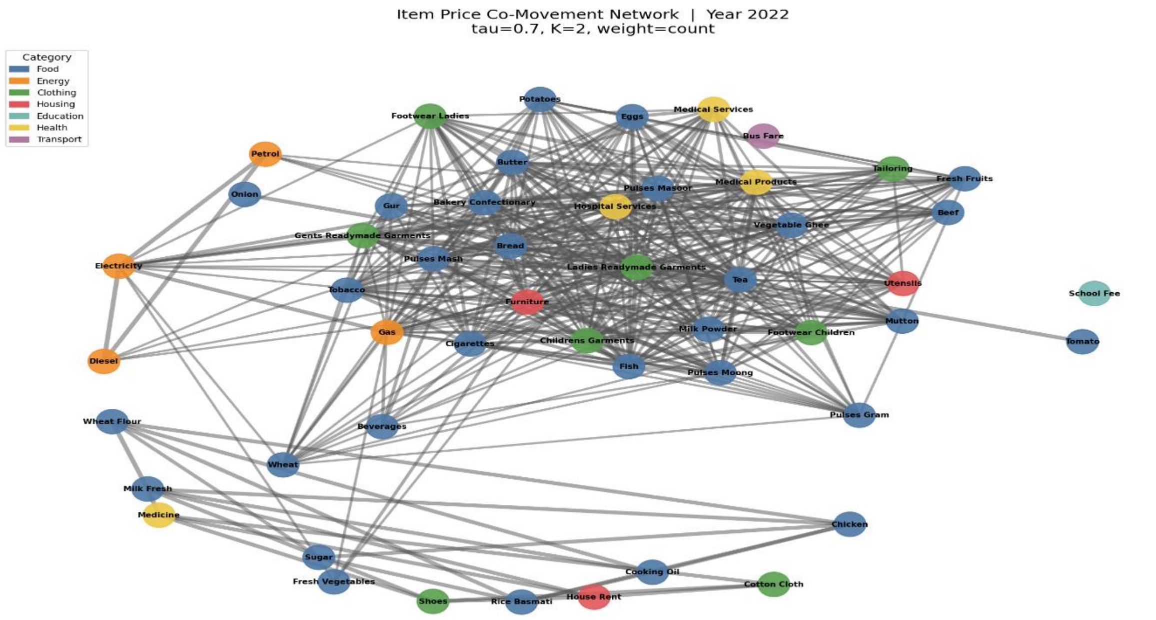
count_similar_cities()	Counts cities where similarity meets or exceeds the threshold tau
build_graph()	Builds NetworkX graph with nodes as items and edges where city count meets K
compute_centrality()	Returns degree, closeness, and betweenness centrality dictionaries
sensitivity_analysis()	Runs the pipeline over multiple tau and K combinations and reports edge counts
temporal_analysis()	Compares edge sets and top central items across all three yearly graphs
category_analysis()	Counts within-category and between-category edges

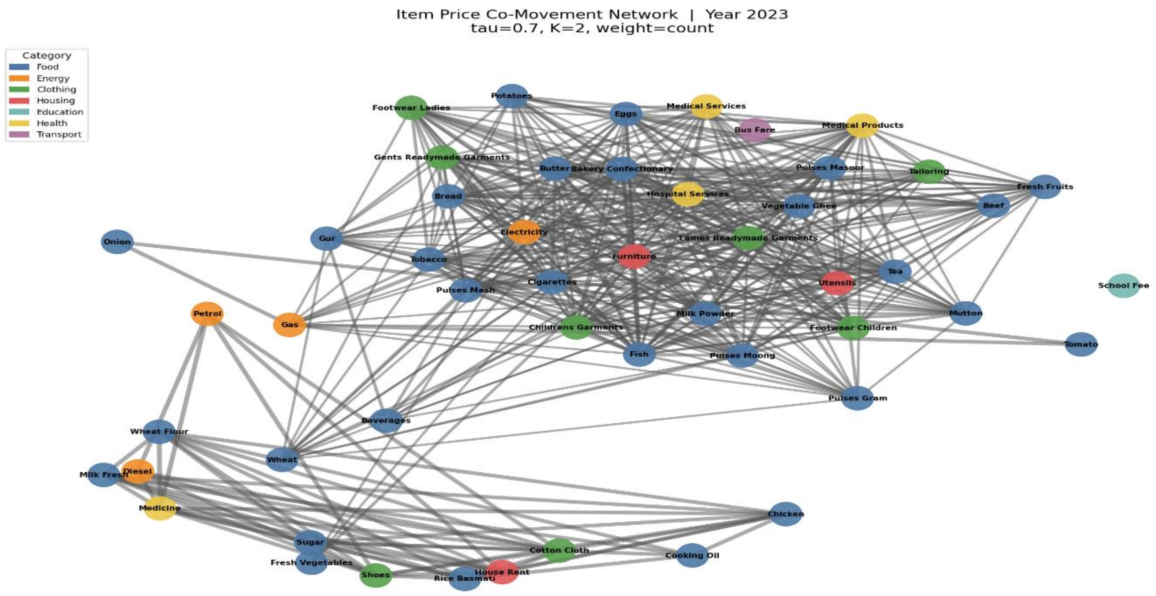
All visualization functions save PNG files to the working directory. The script generates: one graph image per year, a side-by-side three-year comparison, a threshold comparison, a weighting scheme comparison, and centrality bar charts.

5. Graph Construction Results

Using tau = 0.7 and K = 2, the three yearly graphs have the following basic properties:

Year	**Nodes**	**Edges**
2022	51	409
2023	51	431
2024	51	471


Item Price Co-Movement Network | Year 2022
tau=0.7, K=2, weight=count
Category
Food
Energy
Clothing
Housing
Education
Health
Transport
Potatoes
Footwear Ladies
Eggs
Medical Services
Bus Fare
Petrol
Butter
Tailoring
Fresh Fruits
Onion
Pulses Masoor
Medical Products
Gur
Bakery Confectionary
Hospital Services
Beef
Vegetable Ghee
Gents Readymade Garments
Bread
Pulses Mash
Electricity
Ladies Readymade Garments
Tea
Utensils
School Fee
Tobacco
Furniture
Mutton
Milk Powder
Footwear Children
Gas
Cigarettes
Childrens Garments
Tomato
Diesel
Fish
Pulses Moong
Pulses Gram
Wheat Flour
Beverages
Wheat
Milk Fresh
Medicine
Chicken
Sugar
Cooking Oil
Fresh Vegetables
House Rent
Cotton Cloth
Shoes
Rice Basmati


Item Price Co-Movement Network | Year 2023
tau=0.7, K=2, weight=count
Category
Food
Energy
Clothing
Housing
Education
Health
Transport
Potatoes
Footwear Ladies
Eggs
Medical Services
Bus Fare
Medical Products
Gents Readymade Garments
Butter
Bakery Confectionary
Pulses Masoor
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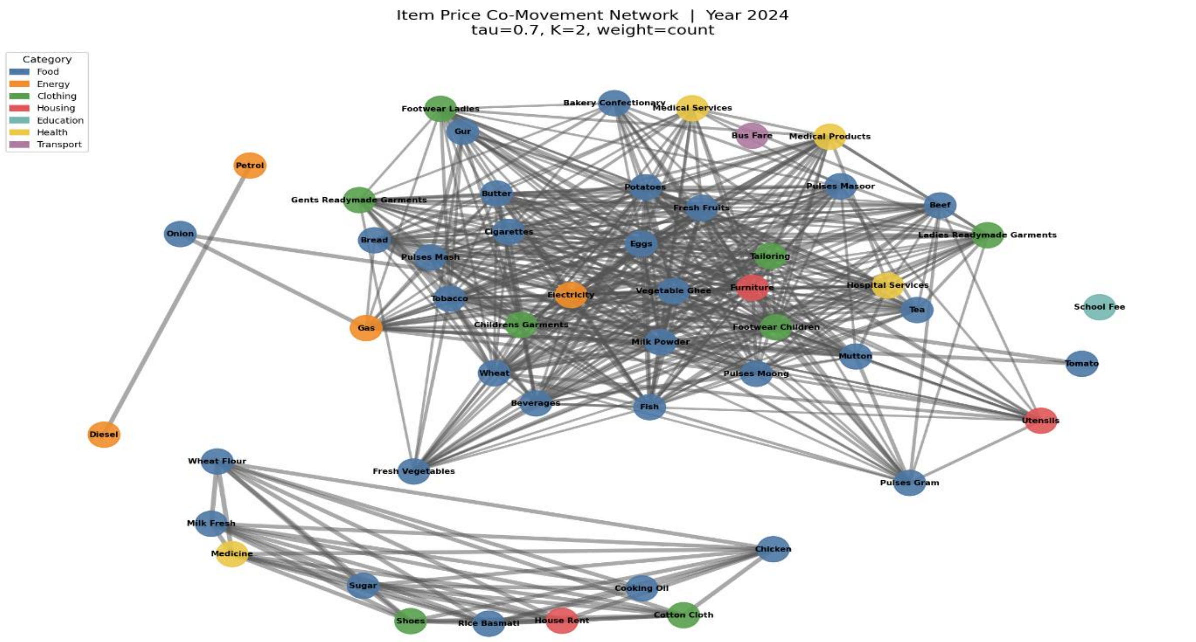
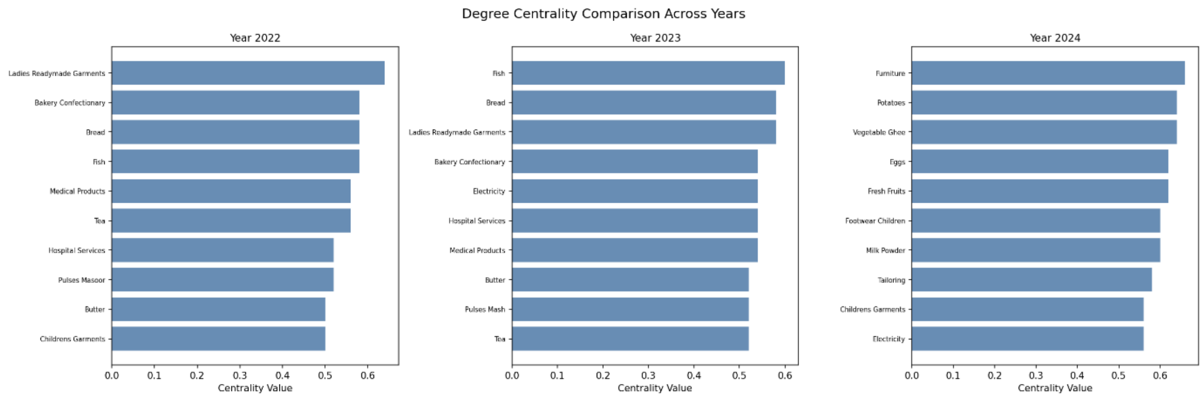


The number of nodes stays fixed at 51 because we include all items as nodes no matter whether they have edges. The edge count grows each year, from 409 in 2022 to 471 in 2024, which shows that price movements across different items became more in sync over time. This matches the economic conditions in Pakistan, where broad inflationary pressures in 2023 and 2024 pulled prices of many different items upward together.

The graph density also increases from about 0.32 in 2022 to 0.37 in 2024. A network with higher density has more connected items, which can mean that price increases in one category tend to happen at the same time as increases in other categories as well.

6. Graph Analysis and Centrality Results

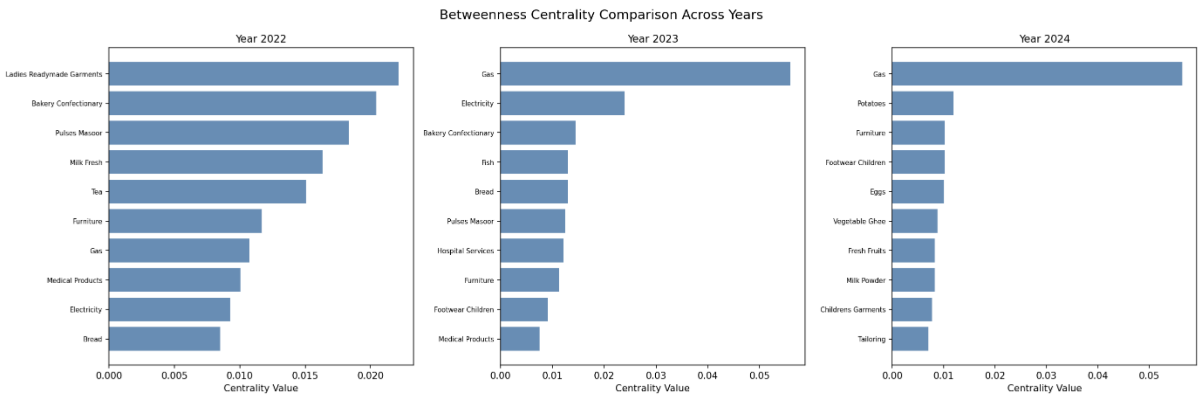


6.1 Year 2022

In 2022, the network had 409 edges. The top item by all three centrality measures was Ladies Readymade Garments. This is somewhat unexpected for a clothing item, but it makes sense in the context of 2022 when supply chain problems and rising cotton and textile input costs drove clothing prices up in patterns that were similar to food and energy items across all five cities.

The top five items by degree centrality in 2022 were Ladies Readymade Garments (0.640), Bakery Confectionary (0.580), Bread (0.580), Fish (0.580), and Medical Products (0.560). These items are highly connected, meaning their price movements tend to move in sync with the largest number of other items in the basket.

For betweenness centrality, Ladies Readymade Garments (0.022) and Bakery Confectionary (0.020) rank highest, followed by Pulses Masoor (0.018), Milk Fresh (0.016), and Tea (0.015). High betweenness means these items act as bridges between otherwise disconnected groups of items in the network.



6.2 Year 2023

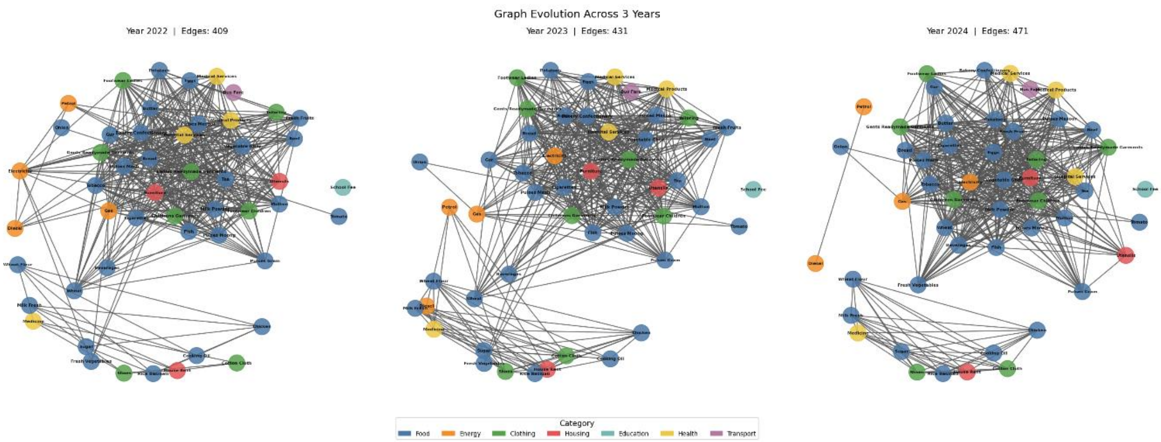
In 2023, the top item by degree and closeness centrality shifted to Fish (0.600), followed by Bread (0.580) and Ladies Readymade Garments (0.580). This change reflects the economic problems of 2023 in Pakistan, when food prices spiked sharply due to floods, import restrictions, and rupee devaluation, creating strong co-movement among staple food items.

A notable observation for 2023 is the rise of Gas in betweenness centrality. Gas scored 0.056 in betweenness, well above any other item. Gas prices in Pakistan changed a lot in 2023 following the removal of subsidies, and the resulting price shock seemed to bridge multiple categories, connecting energy items with food and housing.

6.3 Year 2024

By 2024, the network had become even denser with 471 edges. The top item by degree centrality was Furniture (0.660), followed by Potatoes (0.640), Vegetable Ghee (0.640), Eggs (0.620), and Fresh Fruits (0.620). The presence of food items at the top is in line with ongoing price increases throughout 2024.

Gas again held the top position for betweenness centrality in 2024 with a score of 0.056, showing its role as a connecting bridge in the price co-movement network across all three years.



7. Temporal Network Analysis

7.1 Edge Stability

Out of all edges that appear across the three years, 284 edges are present in all three yearly graphs. These stable edges stand for item pairs that keep moving together in price no matter what the specific economic conditions of any given year are. A few examples of these lasting relationships are:

- Diesel and Petrol: both fuel items track almost perfectly with each other in price every year, which makes logical sense since they are linked by global oil prices.
- Onion and Tomato: both are seasonal vegetables with similar harvest cycles across Pakistani cities.
- Chicken, Cooking Oil, Milk Fresh, Rice Basmati, Sugar, and Wheat Flour: these core food items form a stable cluster, connected to each other across all three years.

- House Rent, Medicine, and Cotton Cloth: a cluster of household cost items that move together.

From 2022 to 2023, the network gained 46 new edges while losing 24. The net addition suggests that price co-movement grew in 2023. From 2023 to 2024, the changes were larger: 153 new edges appeared while 113 disappeared. This higher change in edge structure reflects the more unstable economic environment of 2024.

7.2 Stability of Central Items

Gas consistently ranks at or near the top of betweenness centrality across all three years, acting as a bridge between energy items and the rest of the consumption basket. Bread appears in the top five for degree centrality in all three years, reflecting its central role as a staple whose price movements tend to relate with a wide range of other goods. Ladies Readymade Garments was dominant in 2022, dropped slightly in 2023, and remained active in 2024.

8. Category-Based Analysis

Each item belongs to one of seven set categories. We analyzed whether edges in the network tend to occur within categories or between categories.

Year	Within-Category Edges	Between-Category Edges
2022	176 (43.0%)	233 (57.0%)
2023	177 (41.1%)	254 (58.9%)
2024	199 (42.3%)	272 (57.7%)

In all three years, most edges are between categories rather than within categories. This tells us that price co-movement in Pakistan is not limited to items that are economically similar. A food item like Bread can move in sync with a health item like Medical Products or a clothing item like Ladies Readymade Garments. This cross-category pattern points to broad large economic shocks, such as currency devaluation or fuel price increases, that push prices upward at the same time across unrelated product groups.

Within the Food category, the strongest internal clusters are between Pulses of different types (Gram, Mash, Masoor, Moong), between cooking staples (Rice, Sugar, Wheat Flour, Cooking Oil), and between fresh produce items (Onion, Tomato, Fresh Vegetables, Potatoes). The Energy category shows a tight link between Petrol and Diesel.

9. Sensitivity Analysis

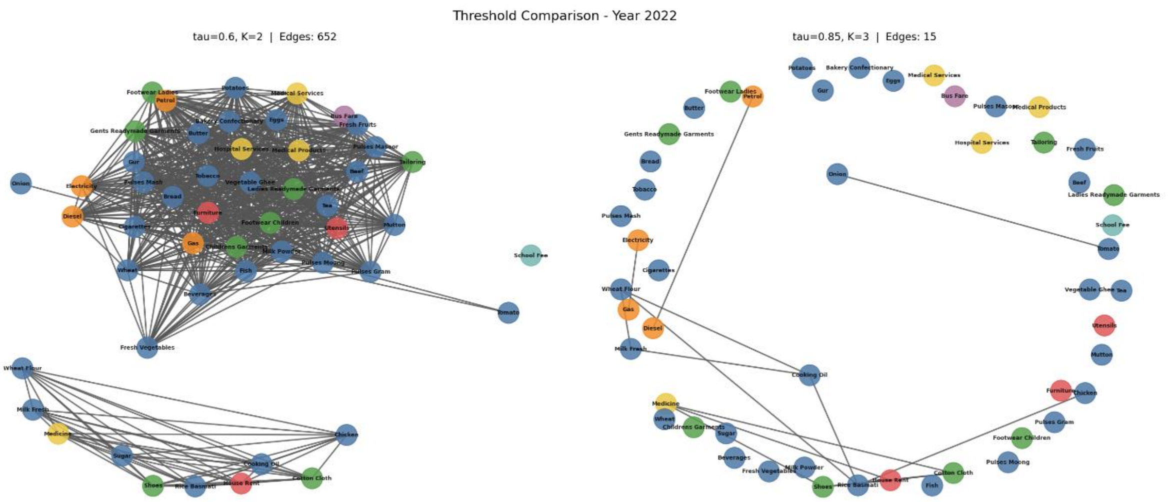
We tested the effect of changing the similarity threshold tau and the city-count threshold K on the number of edges formed in the Year 2022 graph. Results are shown below:

tau	**K**	**Edges in 2022 Graph**
0.5	1	759
0.5	2	716
0.5	3	667
0.6	1	724
0.6	2	652
0.6	3	508
0.7	1	605
0.7	2	409
0.7	3	190
0.8	1	347
0.8	2	116
0.8	3	32
0.9	1	50
0.9	2	11

0.9	3	10

As expected, the number of edges goes down when we raise either tau or K. At tau = 0.5 and K = 1, nearly all possible item pairs form an edge, creating an almost complete graph. At tau = 0.9 and K = 3, only 10 edges survive, standing for only the most solid and consistent price co-movement relationships across all five cities.

The sensitivity analysis confirms that our main parameter choice of tau = 0.7 and K = 2 produces a graph with a fair number of edges that is neither too sparse to be meaningful nor too dense to be informative. The choice of weighting scheme (city count vs average similarity) does not change the edge structure but does affect the edge thickness in visualizations. Items connected to many cities with moderate similarity versus items connected to fewer cities with very high similarity appear differently under the two schemes.



10. Conclusion

This project successfully built and analyzed an item-level price co-movement network from real Consumer Price Index data for Pakistan. The main findings are:

- Price co-movement networks in Pakistan became more densely connected from 2022 to 2024, with edges increasing from 409 to 471 at the same threshold settings. This shows increasing broad price coordination.

- Gas consistently acts as a bridge across different product categories, connecting energy items with food, clothing, and housing items through high betweenness centrality in all three years.
- Most item connections in the network are between categories rather than within categories, confirming that broad economic shocks drive cross-sector price correlation.
- 284 edges were stable across all three years, standing for structural relationships in the consumption basket such as fuel pairs, staple food clusters, and housing cost items.
- Sensitivity analysis showed that the network structure is solid under moderate changes in parameters and that the choice of tau = 0.7 and K = 2 provides a balanced and easy-to-read graph.

The approach can be extended in future work by adding more cities, using weighted centrality measures, applying community-finding methods to identify tighter clusters, or building a temporal network model that tracks edge changes non-stop rather than year by year.

11. References

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YouTube video Link: https://youtu.be/W0qRFxheudQ

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